

# Supporting Information

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## Auditing an AI-Agent Computational Drug-Discovery Workflow Through an SMPD1 Case Study

Yiming Zuo

School of Chemistry, Chemical Engineering and Life Sciences, Wuhan University of Technology, Wuhan 430070, China

### SI-1: Complete evidence matrix (22-gene panel)

Full weighted scoring table. Source: data/panel\_scores.json, data/ot\_associations.json, data/pubmed\_trends.json, scripts/score\_panel.py.

Weights: genetics/omics 0.30, mechanism 0.20, novelty 0.20, safety 0.20, druggability 0.10.

Columns: gene, total score, genetics, mechanism, novelty, safety, druggability, approved drugs, clinical candidates, notes.

### SI-2: Screening funnel and full ranked output

- Library: data/chembl\_library.jsonl (20,111 ChEMBL entries, CNS filters)
- Filtered: data/library\_filtered.csv (11,758 passing PAINS/Brenk/charge)
- Dock set: data/dock\_set.smi (3,500 scaffold-capped)
- Full ranking: results/hits\_ranked.csv (3,642 rows: ligand, Vina score, properties, novelty)
- Funnel registry: results/funnel\_registry.json
- Preparation failures: data/prep\_failures.json (10 embed-failures)

### SI-3: Analog expansion results

225 analogs (F/Cl/OH/OMe/CN/CF<sub>3</sub>/Me at aromatic C–H positions of 6 priority hits), all prepped, docked, 0 failures.

results/expansion\_results.json

### SI-4: ADMET and hERG/DILI predictions

- DeepPurpose ADMET (BBB, P-gp, CYP3A4/2D6, Caco-2, HIA, ClinTox) for top 20: results/admet\_predictions.csv
- Self-trained RF500 hERG (655-mol set) + RF500 DILI (475-mol set): results/herg\_dili\_predictions.csv
- hERG\_Central RF300 reproducibility package: results/herg\_central\_repro/ (dataset SHA256, split indices, metrics, candidate overlap)

### SI-5: MD trajectories and analyses

- Apo frozen-pocket MD: md/prod.dcd (1,161 frames in the audited trajectory inventory), md/solvated.pdb, md/log.csv
- Bound v1 (Li-2013 12-6 Zn): md/bound.dcd, md/bound\_log.csv — **superseded by v2**
- Bound v2 (12-6 subset of 12-6-4): md/bound\_v2.dcd, md/bound\_v2\_log.csv, md/bound\_ref.pdb, md/bound\_analysis\_pre\_registered.json

- Pocket-residence analysis: runs/audit-20260913/tasks/sim-02/attempt-1/pocket\_residence.json
- Snapshots: md/snapshot\_{1..5}ns\_a1.pdb (crystal-aligned)
- MD ensemble docking: results/md\_ensemble\_docking.csv

## SI-6: Audit trail

- Round-1 audit: independent-audit/REPORT.md + 5 verification JSONs
- Round-3 re-review: independent-audit/rereview-r3/REPORT.md + checks
- Round-4 adjudication: independent-audit/round4-acceptance/adjudication.md
- Round-11 acceptance: independent-audit/round11-acceptance/round11-acceptance.md
- Final acceptance: independent-audit/final-acceptance/final-acceptance.md + final\_checks.json
- Verdict-authority resolution: independent-audit/round4-acceptance/verdict-authority-status.json (superseded by adjudication)

## SI-7: Seeded-error benchmark

- 18 seeded cases across 9 error classes, plus the recorded controls
- Expected outcomes frozen in scripts/benchmark\_full.py before that development run; the cases and checks were designed within the same project and were not independently preregistered or held out
- Results: runs/benchmark-full/benchmark\_full\_results.json
- Real-findings replay (13 executed): runs/benchmark-seeded/real\_findings\_replay\_executed.json
- Miss analysis: C4-b (numeric-only check cannot catch citation mismatch), C6-a (atom-name filter cannot catch residue-name contamination), C8-c (naive numbering cannot catch mapping errors). C9-b exposed a control-label/semantics defect; no zero-false-positive claim is made.

## SI-8: Frozen integrity manifest

audit\_manifest.json: 5-level leaf-to-root SHA256 manifest (L1 evidence, L2 scripts, L3 reports, L4 nested handoffs, L5 audit artifacts). No self-reference. 77 entries. research-log.md frozen at D23 state.

## SI-9: GNINA CNN rescoring

9 poses (6 candidates + 3 references). CNNscore and CNNaffinity table. Source: results/gnina\_rescoring.json.

## SI-10: P2Rank pocket prediction

Top pocket: score 39.05, probability 0.961 (6× runner-up), centers overlap with docking box. Source: results/p2rank\_out/.

## SI-11: Model separation

MODEL-A (RF500 full-data legacy) vs MODEL-B (RF300 held-out split) — candidate probabilities side-by-side. Source: results/herg\_central\_repro/models/candidate\_predictions\_by\_model.csv, results/herg\_central\_repro/models/model\_provenance.json.

## **SI-12: Homolog selectivity**

Max Tanimoto vs SMPD3/SMPD2/SMPD4/ASAH1/ASAH2/NAAA ligand sets  $\leq 0.31$  for all top-20. Source: results/homolog\_selectivity.csv.

**All SI items are included in the project directory and are individually hash-tracked via audit\_manifest.json.**